

WR-696 Watershed Problem Analysis

Lynette Schwanger

Greenhouse Gas Dynamics in Restored Wetlands

Introduction

Wetlands are transition zones between land and water where surface water or a high water table saturates the ground often enough and long enough to support hydrophytes and create hydric soils with slowed decomposition and organic matter buildup. These ecosystems play an important role in the global carbon cycle, as they have the potential to store and sequester significant quantities of soil organic carbon (SOC) over time. The hydrology of wetlands limits oxygen availability, slowing decomposition and allowing organic matter buildup. Through SOC accumulation and carbon dioxide (CO₂) uptake of vegetation, wetlands can be significant carbon sinks. However, wetlands can also be a source of methane (CH₄) and nitrous oxide (N₂O). These greenhouse gases (GHGs) have 80 and 263 times the global warming potential of CO₂ on a 20-year time horizon, respectively (Intergovernmental Panel on Climate Change, 2023). As a result, the climate value of wetland conservation and restoration depends not only on carbon storage and CO₂ exchange, but also on the level of CH₄ and N₂O emissions.

In recent years, wetland restoration has been explored as a potential climate mitigation strategy, yet the dynamics of restored wetlands and their relationship to greenhouse gas emissions is poorly understood (Valach et al., 2021; Schuster et al., 2025). Identifying the conditions under which restored wetlands deliver net climate benefits remains a central challenge for restoration planning. GHG emissions from restored wetlands are highly variable, ranging from net climate sinks to net sources. Differences in starting conditions, water table level and

degree of saturation, past land use, restoration techniques, management, vegetation, temperature, salinity, and time since restoration all affect a restored wetland's status as either a GHG sink or source. This review aims to synthesize wetland characteristics and restoration approaches that research has linked to changes in SOC and in CO₂, CH₄, and N₂O emissions to better inform restoration decisions from a climate mitigation perspective.

Background

Wetlands occupy a relatively small fraction of the Earth's land surface, estimated at 5-8% globally, yet they hold a disproportionately large share of terrestrial carbon (Mitsch et al., 2013). The carbon stored in freshwater wetland soils constitutes roughly one-third of the world's total soil carbon stocks, and freshwater wetlands have the potential to serve as important long-term carbon sinks (Schuster et al., 2025). This storage capacity reflects the role of anaerobic soil conditions in slowing microbial decomposition, allowing organic matter to accumulate over centuries to millennia (Shuster et al., 2024). In North America alone, terrestrial wetlands cover approximately 2.2 million km² and hold an estimated 161 Pg C, with peatlands accounting for the majority of that stock (Kolka et al., 2018). These carbon dynamics place wetlands at the center of debates about natural climate solutions, though their role in the global GHG budget is complicated by the fact that wetlands also emit 20-25% of global methane emissions (Mitsch et al., 2013).

Despite their ecological and climatic importance, wetlands have experienced severe and widespread loss over the past three centuries. An estimated 21% of the area of global freshwater wetlands was lost to drainage and land-use conversion between 1700 and 2020 (Fluet-Chouinard et al., 2023). In North America, the history of wetland loss is closely tied to agricultural

expansion, with wetlands long regarded as unproductive land requiring drainage before they could be put to productive use (Kolka et al., 2018). The drainage of organic soils for agriculture is particularly consequential for the carbon cycle, as it exposes previously anaerobic peat to oxygen, triggering rapid oxidation and sustained CO₂ emissions that can persist for decades. In the Sacramento-San Joaquin Delta, for example, drainage converted a roughly 6,700-year-old carbon sink to a CO₂ source, producing ongoing emissions estimated to account for 21% of California's annual agricultural GHG output (Deverel et al, 2020; Drexler et al, 2009). Globally, the degradation of natural wetlands has effectively reversed their carbon sink function in many regions, turning ecosystems that historically removed carbon from the atmosphere into net sources of CO₂ and CH₄ (Schuster et al., 2025).

Recognition of these losses has driven growing interest in wetland restoration as a natural climate solution. In the United States, a policy of “no net loss” of wetlands adopted in 1989 marked a turning point, dramatically slowing wetland conversion and establishing the framework for wetland mitigation banking programs that offset development-driven losses with restored or created wetlands (Kolka et al., 2018). More recently, restored wetlands have been increasingly considered as part of broader climate mitigation strategies, alongside the preservation of intact ecosystems such as forests and coastal blue carbon systems. Wetland restoration holds appeal as a climate solution not only because of its carbon sequestration potential, but because it simultaneously addresses multiple environmental goals, including water quality improvement, flood attenuation, drought resilience, and habitat for native species (Chausson et al., 2020; Thorslund et al., 2017). However, whether any given restoration effort delivers a net climate benefit depends on a complex interplay of GHG fluxes, not just carbon sequestration alone. Restored wetlands that increase CH₄ and N₂O emissions can offset or even reverse the climate

gains from CO₂ uptake and soil carbon accumulation (Pfeifer-Meister et al., 2018; Shuster et al., 2024). One framework for evaluating this tradeoff is how long after restoration it takes for cumulative CO₂ sequestration to outweigh the cumulative radiative forcing from elevated CH₄ and N₂O emissions. Understanding the site-level factors that shorten or lengthen that window is therefore central to evaluating wetland restoration as a practical climate intervention.

Influences on CO₂ Emissions and SOC Accumulation

Water table level is the most consistent predictor of both SOC accumulation and CO₂ exchange in restored wetlands, largely because it controls how much oxygen is available for microbial decomposition. When soils are saturated, anaerobic conditions slow organic matter breakdown and allow carbon to accumulate, whereas even moderate drawdowns of the water table can trigger rapid oxidation of carbon that was previously stored in the wetland's soils. In the Sacramento-San Joaquin Delta in California, restored wetlands maintained under flooded conditions were strong net CO₂ sinks, sequestering between 315 and 358 g C m⁻² yr⁻¹, while adjacent drained agricultural peatlands released up to 341 g C m⁻² yr⁻¹ as CO₂, driven by high rates of peat oxidation and ecosystem respiration (Knox et al., 2015). Photosynthesis was similar across flooded and drained sites, but flooding suppressed ecosystem respiration enough to tip the balance toward net CO₂ uptake (Knox et al., 2015). In long-term eddy covariance records from restored wetlands in the Sacramento-San Joaquin Delta, water table depth was consistently one of the dominant environmental controls on net ecosystem exchange, accounting for roughly 10-30% of explained variance across mature sites (Kasak et al., 2026). Together, these findings suggest that the ability of a restored wetland to function as a CO₂ sink is closely tied to whether water table elevation is sufficient to sustain anaerobic soil conditions across seasons.

Previous land use and site history shape the starting conditions for SOC recovery in ways that can persist for decades after restoration. Wetlands that were drained and cultivated for agriculture typically enter restoration with depleted soil carbon stocks, altered microbial communities, and reduced availability of fresh organic substrates, all of which slow the return of carbon-sequestering function. In Louisiana, naturally occurring wetlands had higher SOC accumulation rates than restored sites on average, though restored sites showed increasing rates over time that were correlated with wetland age, suggesting that recovery is ongoing but slow (Suir et al., 2019). The study also found that carbon accumulation rates were not uniform across restored site types, with hydrogeomorphic setting and particularly river connectivity having a strong influence on which sites accumulated carbon most rapidly. Sites in basins with greater riverine sediment and nutrient inputs, such as the Mississippi River and Atchafalaya basins, had significantly higher accumulation rates than more isolated coastal sites, highlighting the importance of broader landscape context alongside site-level restoration history.

Vegetation establishment plays a central role in driving CO₂ uptake after restoration, and the pace at which plants colonize a site has a measurable influence on when a restored wetland transitions to a net carbon sink. In a multi-site eddy covariance study of five restored Delta wetlands, vegetation cover explained roughly half of the interannual variability in annual net ecosystem exchange, with a minimum threshold of approximately 55% cover needed before sites became consistent net CO₂ sinks (Valach et al., 2021). Sites that achieved rapid vegetation establishment, typically within two years, were more reliably stable CO₂ sinks, while sites with slower colonization rates had higher interannual variability and occasional source years.

Restoration design factors such as water depth and planting strategy were found to strongly influence vegetation establishment trajectories. Shallower water conditions and

transplanted patches of emergent macrophytes promoted faster colonization than deeper, unplanted areas, suggesting that design decisions made at the outset of restoration have lasting effects on carbon uptake potential (Valach et al., 2021; Kasak et al., 2026). The type of plant litter also matters for long-term SOC accumulation. In a paired study of restored and degraded riparian wetlands in Australia, native wetland species produced leaf litter that decomposed more slowly in the soil than litter from invasive species present in degraded sites, and this difference was associated with greater SOC preservation in the restored wetlands (Schuster et al., 2025). This finding implies that the carbon balance of restored wetlands is affected by plant species decomposition rate. SOC stocks in those restored wetlands increased by 12% within one year and 53% within six years of restoration, demonstrating that vegetation-driven SOC gains can accumulate relatively quickly when plant communities are re-established (Schuster et al., 2025).

Environmental conditions, including salinity, temperature, and nutrient availability, modify both the rate of SOC accumulation and the strength of CO₂ exchange in restored wetlands, though their effects are often site-specific and interact with hydrology and vegetation. Salinity influences SOC dynamics primarily through its effects on decomposition and microbial community structure. In coastal and estuarine systems, higher salinity tends to suppress microbial activity and organic matter breakdown, which can promote carbon preservation in sediments. In Louisiana coastal wetlands, higher-salinity zones had significantly higher carbon accumulation rates than intermediate or brackish zones, consistent with reduced mineralization of organic matter under saline conditions (Suir et al., 2019). Temperature also shapes CO₂ and SOC dynamics by regulating both photosynthesis and respiration, and in warm climates with long growing seasons, the net effect tends to favor high CO₂ uptake. Eddy covariance data from the Sacramento-San Joaquin Delta suggest that the region's long, warm growing season is part of

why restored Delta wetlands rank among the strongest CO₂ sinks globally, with mean annual CO₂ uptake across mature sites of $-449 \pm 173 \text{ gC m}^{-2} \text{ yr}^{-1}$ (Kasak et al., 2026). Nutrient availability can accelerate vegetation establishment and productivity, which can benefit CO₂ uptake in the short term, but high phosphorus conditions in particular have been shown to promote low-diversity plant communities dominated by fast-growing species that produce more labile litter and may ultimately reduce organic inputs to the soil (Valach et al., 2021).

Time since restoration is an important but uneven predictor of CO₂ sink strength and SOC accumulation, with some functions recovering relatively quickly and others requiring decades. For CO₂ exchange, the transition to net carbon uptake can be rapid in productive systems. Most restored wetlands in the California Delta became net CO₂ sinks within two years of re-flooding once vegetation, primarily tule and cattails (*Schoenoplectus acutus* and *Typha*), exceeded the 55% cover threshold (Valach et al., 2021). A riparian wetland restoration project in Australia where sites had a variety of vegetation including native trees, and native and non-native shrubs, herbs and graminoids, produced measurable reductions in net carbon emissions within a single year and sustained SOC gains over six years (Schuster et al., 2025). In subtropical coastal wetlands, nine years after tidal reintroduction was sufficient to generate significant GHG mitigation through woody biomass growth, soil accretion, and reduced N₂O emissions from the cessation of fertilization (Iram et al., 2022). However, the pace of recovery depends heavily on whether early conditions favor vegetation establishment, and sites with delayed or incomplete vegetation colonization can remain net CO₂ sources for years. Long-term eddy covariance records from the Delta show that sites with persistent vegetation establishment failures remained GHG sources even many years after restoration began, while those with rapid recolonization achieved earlier transitions to net sink status (Kasak et al., 2026). For SOC accumulation

specifically, the chronosequence analysis of Louisiana coastal wetlands found that carbon accumulation rates increased progressively with restoration age, but most restored sites required decades before approaching the carbon storage of natural reference wetlands (Suir et al., 2019). This suggests that while CO₂ exchange can recover on relatively short timescales in favorable conditions, SOC stock recovery is slower and may not converge with natural wetland conditions within management-relevant timeframes.

Influences on CH₄ Emissions

As with CO₂, water table depth is the most consistent control on methane emissions in restored wetlands because it regulates the oxygen conditions that either suppress or promote methanogenesis. Methanogenesis is a form of anaerobic respiration performed exclusively by specialized microbes known as methanogens, which create methane as a metabolic byproduct. Across the literature, rewetting generally increases CH₄ production by creating more anaerobic soil conditions, but the relationship is not always linear. In a recent global meta-analysis, rewetting was identified as one of the dominant predictors of methane responses after restoration, and rewetting to water tables shallower than -15 cm increased CH₄ emissions in restored wetlands (Xu et al., 2025). At the site scale, water level effects have also been described as important but complex, with restoration flooding causing an initial increase in methane emissions, but with the highest CH₄ fluxes occurring when the water level was near the soil surface rather than under deeper inundation (Rey-Sanchez et al., 2025). This suggests a threshold-type pattern where methane emissions increased as water levels rose from belowground toward the surface, but once water levels were well above the surface, fluxes sometimes declined, likely because deeper water columns increased opportunities for methane

oxidation and reduced ebullition, or the bubbling up, of CH₄ to the surface (Rey-Sanchez et al., 2025). A similar hydrologic sensitivity appears in the Prairie Pothole Region, where CH₄ emissions peaked when water-filled pore space exceeded 60%, while negative methane fluxes occurred as soils became drier and water-filled pore space approached 40% (Gleason et al., 2009). Together, these studies suggest that restoration-driven increases in CH₄ are most likely when hydrologic conditions keep soils saturated or nearly saturated, but not necessarily under the deepest or most prolonged flooding.

Salinity is also a major control on methane emissions, though its effects vary by wetland type and salinity range. Broadly, the literature suggests that higher salinity tends to suppress CH₄ production because sulfate-rich saline conditions favor microbial pathways that compete with methanogenesis. This pattern is strongly supported by a global meta-analysis, which found that saline wetlands showed no significant change in CH₄ emissions after restoration. In contrast, freshwater wetlands, peatlands, and seasonally flooded wetlands had significant increases in methane emissions after restoration (Xu et al., 2025). Across four tidal wetlands, methane emissions also decreased substantially as salinity increased. Lower-salinity wetlands had more labile organic carbon and nutrients, which favored methanogenesis, whereas higher salinity depressed decomposition and microbial activity, leading to much lower CH₄ production in saline wetlands (Ho et al., 2024). A similar estuarine pattern has been reported in restored and reference wetlands, where CH₄ fluxes were highly variable and sometimes high under low saline conditions, but relatively low on an annual basis at salinities above 2 ppt. This suggests that estuarine wetlands with low salinity are more likely to have elevated methane emissions that may offset carbon sequestration benefits (Williams et al., 2025). However, salinity is not always a simple predictor at the site level. In one restored wetland, salinity did not clearly reduce methane

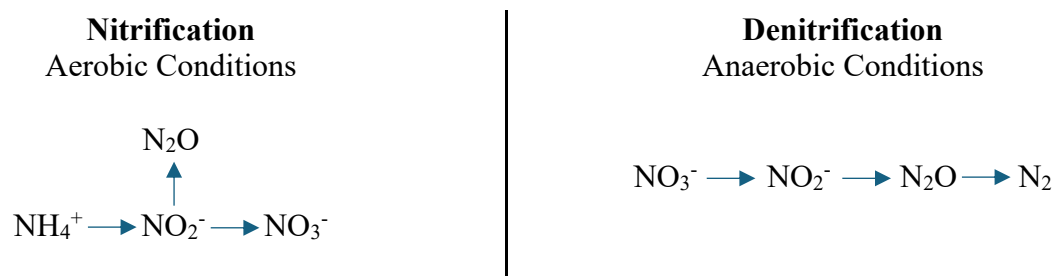
emissions even when summer salinity exceeded 5 ppt, and late-summer declines in CH₄ could not be confidently attributed to salinity alone because salinity and water-level decline occurred at the same time (Rey-Sanchez et al., 2025). Overall, salinity seems to be one of the main factors limiting methane emissions, but its influence may depend on whether salinity crosses key thresholds and on how it interacts with hydrology and porewater chemistry.

Temperature is often thought to be an important driver of methane emissions because warmer conditions increase microbial activity, including methanogenesis. However, the literature suggests that its effect is often secondary to water table depth and site-based studies show that warmer conditions do not always translate neatly into higher methane emissions. Seasonal cycles of both CH₄ and CO₂ have been shown to be only partially driven by soil temperature, since flux peaks did not always coincide with peaks in soil temperature (Rey-Sanchez et al., 2025). In a restored headwater riverine wetland, temperature helped explain variability in dissolved CH₄, but its role depended on context. CH₄ variability was better explained by a combination of temperature, dissolved oxygen, and other water-quality variables, and there was even a negative relationship between water temperature and CH₄ saturation under some conditions, implying that methane production and accumulation were shaped by dissolved oxygen dynamics as much as by warming effects (Zhang et al., 2021). Temperature alone also appears insufficient to explain methane dynamics in other restored wetland systems. At a California restoration site, methane emissions were best explained by latent heat on a day-to-day basis, while seasonal changes were better explained by water table depth and surface water conductivity, suggesting that methane emissions are shaped by multiple interacting wetland conditions rather than temperature alone (Delwiche et al., 2025). Taken together, these studies suggest that temperature matters, but

usually as part of a broader environmental network in which hydrology and salinity more directly structure methane production, oxidation, and transport.

Influences on N₂O Emissions

Nitrous oxide is produced in wetland soils primarily through nitrification and denitrification, two microbial nitrogen-cycling processes. In nitrification, N₂O is released as an intermediate byproduct in the aerobic conversion of ammonium (NH₄⁺) to nitrate (NO₃⁻), while denitrification produces N₂O when oxygen availability interrupts the anaerobic reduction of nitrate to nitrogen gas (N₂).



Both processes occur across a range of soil moisture conditions and together account for an estimated 50-70% of annual N₂O fluxes to the atmosphere (Pfeifer-Meister et al., 2018; Morse & Bernhardt, 2013). Because N₂O is both a potent greenhouse gas and the dominant ozone-depleting substance of the twenty-first century (Ravishankara et al., 2009), understanding how wetland restoration alters these processes has important implications for evaluating the climate costs and benefits of restoration efforts.

Soil saturation and inundation is a primary driver of N₂O dynamics in restored wetlands, largely through its control over soil redox conditions. Studies have established that maximum N₂O emissions from soils typically occur when water-filled pore space ranges from approximately 40-80%, conditions that support both aerobic nitrification and incomplete anaerobic denitrification simultaneously (Pfeifer-Meister et al., 2018). In restored and reference

coastal plain wetlands in North Carolina, N₂O fluxes increased with soil moisture across all soil types (Morse and Bernhardt, 2013). Notably, in more organically enriched, acidic soils of restored and forested wetland plots, nitrification was a more important driver of N₂O flux than denitrification. In contrast, restored wetlands with mineral soils typical of former agricultural fields, incomplete denitrification following rain events was the dominant source of N₂O (Morse & Bernhardt, 2013). These findings underscore that the relationship between soil saturation and N₂O flux is not linear, and that hydrologic restoration can shift the balance between these production pathways in ways that are difficult to predict from soil moisture alone.

Land-use history and the nitrogen loading environment of the receiving watershed shape N₂O emissions from restored wetlands in ways that can either suppress or amplify the effects of hydrology on N₂O emissions. One study found that across natural, converted, and restored wetlands, land use did not have a statistically significant effect on N₂O emissions, though restored and converted wetlands had the lowest soil carbon and nitrogen content due to past agricultural use (Kluber et al., 2014). The biogeochemical processes governing denitrification often lag behind the more rapid recovery of hydrological and photosynthetic function following restoration, because microbial communities require more time to re-establish in altered soil conditions (Kluber et al., 2014). In the Willamette Valley in Oregon, no N₂O fluxes were found in natural, restored, and agricultural seasonal wetland prairies, even during extended periods of saturation, and laboratory tests concluded that denitrification and N₂O production were limited by both nitrate and organic carbon availability (Pfeifer-Meister et al., 2018). Similarly modest N₂O fluxes were reported in Prairie Pothole wetlands, with grazing and haying management practices having little effect on N₂O fluxes from either native or restored wetlands, and with

cumulative annual N₂O fluxes from restored wetlands far below those of adjacent croplands (Finocchiaro et al., 2014).

Where nitrogen loading rates are elevated, often found in wetlands designed to intercept agricultural drainage, N₂O emissions tend to scale with nitrate availability, though still at rates substantially lower than those of the surrounding agricultural landscape. N₂O production and emission measured at three Iowa wetlands receiving nonpoint source nitrate loads across a wide range of concentrations, showed that average N₂O emission rates were strongly correlated with inflow nitrate concentrations, loading rates, and removal rates (Ampe et al., 2025). Despite processing large nitrate loads, these wetlands converted only about 0.5% of removed nitrate into N₂O, with the remainder fully denitrified to harmless N₂ gas. Additionally, N₂O emissions from the inundated wetlands were no higher than those from the cropland they replaced and were well below those of downstream streams receiving comparable nitrogen inputs (Ampe et al., 2025). These findings support the conclusion that continuously inundated wetlands exhibit high nitrate conversion efficiency and relatively low N₂O emission rates compared to either seasonally flooded wetlands or agricultural land, and that widespread restoration of such systems to intercept nitrogen in agricultural watersheds is unlikely to produce a net increase in N₂O emissions.

In rewetted peatlands, soil organic carbon content plays a central role in determining whether a restored wetland acts as an N₂O source or sink. Rewetted peatlands with contrasting SOC contents of 6%, 11%, and 23% repeatedly showed net N₂O uptake from the atmosphere regardless of nitrogen fertilization rate, with the highest consumption rates in the highest SOC soils (Ye & Horwath, 2016). This is consistent with more strongly reduced conditions in carbon-rich soils driving denitrification to completion rather than stalling at N₂O. SOC content was the

dominant predictor of N₂O emissions across these sites, with annual N₂O emissions declining from 8.9 kg ha⁻¹ in the lowest-carbon soils to 1.9 kg ha⁻¹ in the highest, following a consistent inverse relationship regardless of season or fertilization treatment (Ye et al., 2016). These findings suggest that SOC content may serve as a practical indicator for predicting N₂O flux in rewetted peatlands, and that maintaining high water tables to reduce oxygen availability can shift rewetted organic soils from N₂O sources to sinks.

Taken together, the evidence suggests that the effect of wetland restoration on N₂O emissions is highly context-dependent, varying with the nitrogen loading environment, soil organic matter content, and hydrology. Synthesized data from freshwater wetland restoration studies globally found that restored and natural wetlands tended to have lower N₂O emissions than degraded wetlands, likely because rewetting reduces nitrate availability through enhanced plant uptake and complete denitrification under strongly anaerobic conditions (Schuster et al., 2024). The weight of evidence across agricultural, coastal plain, peatland, and prairie wetland systems indicates that permanently or semi-permanently inundated restored wetlands are unlikely to increase N₂O emissions relative to the prior degraded or agricultural land use, particularly where soil carbon availability is sufficient to drive complete denitrification.

Conclusion

This review has synthesized evidence across CO₂ and SOC dynamics, CH₄ emissions, and N₂O fluxes to identify the conditions under which wetland restoration is most likely to deliver net climate benefits. Across all three GHG components, hydrology is the dominant and most consistent control. The position of the water table determines the redox regime of restored soils, which governs the balance between carbon storage and CO₂ release, the rate of

methanogenesis, and the partitioning of nitrogen between complete denitrification and N_2O production. Restoration approaches that rapidly reinstate high water tables and maintain them across seasons are the most effective strategies available for improving the GHG balance of a restored wetland. The evidence is consistent across wetland types and geographies. Prolonged anaerobic conditions favor SOC accumulation, suppress N_2O emissions, and, while they can increase CH_4 production, tend to do so at rates that are eventually offset by CO_2 sequestration in productive systems.

Beyond hydrology, the literature identifies several site characteristics that consistently shape GHG outcomes. Salinity suppresses both CH_4 and, in agriculturally affected systems, N_2O emissions, making coastal and estuarine restoration generally less likely to produce short-term net GHG increases than freshwater systems. Soil organic carbon content is a strong predictor of N_2O dynamics, with high-SOC rewetted soils more reliably functioning as N_2O sinks due to the tendency of carbon-rich, strongly reduced conditions to drive denitrification to completion. Previous land use shapes the starting conditions for all three GHGs. Agricultural history depletes SOC stocks, alters microbial communities, and can leave behind nitrogen legacies that elevate N_2O or CH_4 production early in the post-restoration period. Vegetation establishment is critical for CO_2 dynamics specifically, with the transition to net CO_2 uptake closely tied to the pace and extent of plant colonization. Design decisions at restoration outset such as water depth, planting strategy, and species selection, have measurable, lasting effects on how quickly sites reach productive carbon sink status.

Several important uncertainties remain. The timeline over which restored wetlands transition from net GHG sources to net sinks is highly variable and depends on the interplay of all the factors reviewed here. Long-term monitoring studies, particularly those with multi-decade

records spanning both CO₂ and CH₄, remain rare, and most of the literature on N₂O dynamics in restored wetlands comes from a relatively narrow range of system types. Non-peat freshwater marshes and riparian wetlands are underrepresented in studies relative to peatlands, which limits confidence in applying these findings to other restoration contexts. The role of temperature and climate variability as modifiers of restoration outcomes is also incompletely understood, particularly as warming alters the hydrology and biogeochemistry of restored systems on longer timescales. Despite these gaps, the weight of evidence supports the conclusion that wetland restoration can be an effective climate mitigation strategy when implemented in systems where water table depth is consistently above the soil surface, vegetation recovery is rapid, and the nitrogen loading environment does not overwhelm the denitrification capacity of the soil. Prioritizing restoration design choices that promote these conditions, and maintaining long-term monitoring to detect departures from expected GHG trajectories, should be central to restoration planning from a climate perspective.

References

- Ampe, H. L., Crumpton, W. G., Stenback, G. A., & Green, D. I. S. (2025). Nitrous oxide and methane production and emission in wetlands receiving nonpoint source nitrate loads. *Journal of Environmental Quality*, 54(6), 2048–2060. <https://doi.org/10.1002/jeq2.70081>
- Chausson, A., Turner, B., Seddon, D., Chabaneix, N., Girardin, C. A. J., Kapos, V., Key, I., Roe, D., Smith, A., Woroniecki, S., & Seddon, N. (2020). Mapping the effectiveness of nature-based solutions for climate change adaptation. *Global Change Biology*, 26(11), 6134–6155. <https://doi.org/10.1111/gcb.15310>
- Delwiche, K., Matthes, J. H., Arias-Ortiz, A., Knox, S. H., Oikawa, P., Sturtevant, C., Verfaillie, J., Szutu, D., Keenan, T. F., & Baldocchi, D. (2025). Dynamic methane emissions in a restored wetland: Decadal insights into uncertain climate outcomes and critical science needs. *Agricultural and Forest Meteorology*, 373, 110735. <https://doi.org/10.1016/j.agrformet.2025.110735>
- Deverel, S. J., Dore, S., & Schmutte, C. (2020). Solutions for subsidence in the California Delta, USA, an extreme example of organic-soil drainage gone awry. *Proceedings of the International Association of Hydrological Sciences*, 382, 837–842. <https://doi.org/10.5194/piahs-382-837-2020>
- Finocchiaro, R., Tangen, B., & Gleason, R. (2014). Greenhouse gas fluxes of grazed and hayed wetland catchments in the U.S. Prairie Pothole Ecoregion. *Wetlands Ecology and Management*, 22(3), 305–324. <https://doi.org/10.1007/s11273-013-9331-5>
- Fluet-Chouinard, E., Stocker, B. D., Zhang, Z., Malhotra, A., Melton, J. R., Poulter, B., Kaplan, J. O., Goldewijk, K. K., Siebert, S., Minayeva, T., Hugelius, G., Joosten, H., Barthelmes, A., Prigent, C., Aires, F., Hoyt, A. M., Davidson, N., Finlayson, C. M., Lehner, B., ... McIntyre, P. B.

(2023). Extensive global wetland loss over the past three centuries. *Nature*, 614(7947), 281–286.

<https://doi.org/10.1038/s41586-022-05572-6>

Gleason, R. A., Tangen, B. A., Browne, B. A., & Euliss, N. H. (2009). Greenhouse gas flux from cropland and restored wetlands in the Prairie Pothole Region. *Soil Biology and Biochemistry*, 41(12), 2501–2507. <https://doi.org/10.1016/j.soilbio.2009.09.008>

Ho, L., Pham, K., Barthel, M., Harris, S., Bodé, S., De Vrieze, J., Vermeir, P., Six, J., Boeckx, P., & Goethals, P. (2024). Unravelling CH₄ and N₂O dynamics in tidal wetlands using natural abundance isotopes and functional genes. *Soil Biology and Biochemistry*, 196, 109497.

<https://doi.org/10.1016/j.soilbio.2024.109497>

Intergovernmental Panel On Climate Change (Ipcc). (2023). *Climate Change 2021 - The Physical Science Basis: Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (1st ed.). Cambridge University Press.

<https://doi.org/10.1017/9781009157896>

Iram, N., Maher, D. T., Lovelock, C. E., Baker, T., Cadier, C., & Adame, M. F. (2022). Climate change mitigation and improvement of water quality from the restoration of a subtropical coastal wetland. *Ecological Applications*, 32(5), e2620. <https://doi.org/10.1002/eap.2620>

Kasak, K., Ahmadi, A., Dronova, I., Arias-Ortiz, A., Wang, T., Valach, A. C., Szutu, D., Verfaillie, J., & Baldocchi, D. D. (2026). Interannual Variability in Greenhouse Gas Emissions Challenges Post-Restoration Net Sink Predictions in California Delta Wetlands. *Global Change Biology*, 32(1), e70700. <https://doi.org/10.1111/gcb.70700>

Kluber, L. A., Miller, J. O., Ducey, T. F., Hunt, P. G., Lang, M., & S. Ro, K. (2014). Multistate assessment of wetland restoration on CO₂ and N₂O emissions and soil bacterial communities. *Applied Soil Ecology*, 76, 87–94. <https://doi.org/10.1016/j.apsoil.2013.12.014>

Knox, S. H., Sturtevant, C., Matthes, J. H., Koteen, L., Verfaillie, J., & Baldocchi, D. (2015).

Agricultural peatland restoration: Effects of land-use change on greenhouse gas (CO₂ and CH₄) fluxes in the Sacramento-San Joaquin Delta. *Global Change Biology*, 21(2), 750–765.

<https://doi.org/10.1111/gcb.12745>

Mitsch, W. J., Bernal, B., Nahlik, A. M., Mander, Ü., Zhang, L., Anderson, C. J., Jørgensen, S. E., & Brix, H. (2013). Wetlands, carbon, and climate change. *Landscape Ecology*, 28(4), 583–597.

<https://doi.org/10.1007/s10980-012-9758-8>

Morse, J. L., & Bernhardt, E. S. (2013). Using 15N tracers to estimate N₂O and N₂ emissions from nitrification and denitrification in coastal plain wetlands under contrasting land-uses. *Soil*

Biology and Biochemistry, 57, 635–643. <https://doi.org/10.1016/j.soilbio.2012.07.025>

Pfeifer-Meister, L., Gayton, L. G., Roy, B. A., Johnson, B. R., & Bridgham, S. D. (2018). Greenhouse gas emissions limited by low nitrogen and carbon availability in natural, restored, and agricultural Oregon seasonal wetlands. *PeerJ*, 6, e5465. <https://doi.org/10.7717/peerj.5465>

Ravishankara, A. R., Daniel, J. S., & Portmann, R. W. (2009). Nitrous Oxide (N₂O): The Dominant Ozone-Depleting Substance Emitted in the 21st Century. *Science*, 326(5949), 123–125.

<https://doi.org/10.1126/science.1176985>

Rey-Sanchez, C., Arias-Ortiz, A., Kasak, K., Shortt, R., Szutu, D., Verfaillie, J., Lorenson, T., Liira, M., Somelar, P., Espenberg, M., & Baldocchi, D. (2025). Explaining hot spots of methane flux in a restored wetland: The role of water level, soil disturbance, and methanotrophy. *Environmental Research Letters*, 20(7), 074064. <https://doi.org/10.1088/1748-9326/ade45b>

Schuster, L., Taillardat, P., Macreadie, P. I., & Malerba, M. E. (2024). Freshwater wetland restoration and conservation are long-term natural climate solutions. *Science of The Total Environment*, 922, 171218. <https://doi.org/10.1016/j.scitotenv.2024.171218>

Schuster, L., Trevathan-Tackett, S., Carnell, P., Morris, K., Mole, B., & Malerba, M. E. (2025).

Restoring riparian wetlands for carbon and nitrogen benefits and other critical ecosystem functions. *Journal of Environmental Management*, 391, 126433.

<https://doi.org/10.1016/j.jenvman.2025.126433>

Suir, G. M., Sasser, C. E., DeLaune, R. D., & Murray, E. O. (2019). Comparing carbon accumulation in restored and natural wetland soils of coastal Louisiana. *International Journal of Sediment Research*, 34(6), 600–607.

<https://doi.org/10.1016/j.ijsrc.2019.05.001>

Tang, W., Krauss, K., Bansal, S., Drexler, J., Wickland, K., Chimner, R., Hogan, D., Pindilli, E. J., Benschoter, B., Tangen, B., Kane, E., Bridgham, S., & Richardson, C. (2018). *Recommended Citation for Chapter*.

Thorslund, J., Jarsjo, J., Jaramillo, F., Jawitz, J. W., Manzoni, S., Basu, N. B., Chalov, S. R., Cohen, M. J., Creed, I. F., Goldenberg, R., Hylin, A., Kalantari, Z., Koussis, A. D., Lyon, S. W., Mazi, K., Mard, J., Persson, K., Pietro, J., Prieto, C., ... Destouni, G. (2017). Wetlands as large-scale nature-based solutions: Status and challenges for research, engineering and management.

Ecological Engineering, 108, 489–497. <https://doi.org/10.1016/j.ecoleng.2017.07.012>

Valach, A. C., Kasak, K., Hemes, K. S., Anthony, T. L., Dronova, I., Taddeo, S., Silver, W. L., Szutu, D., Verfaillie, J., & Baldocchi, D. D. (2021). Productive wetlands restored for carbon

sequestration quickly become net CO₂ sinks with site-level factors driving uptake variability.

PLOS ONE, 16(3), e0248398. <https://doi.org/10.1371/journal.pone.0248398>

Williams, Trevor, Janousek, C. N., McKeon, M. A., Diefenderfer, H. L., Cornu, C. E., Borde, A. B., Apple, J., Brophy, L. S., Norwood, M., Schultz, M. A., & Bridgham, S. D. (n.d.). *Methane and*

nitrous oxide fluxes from reference, restored, and disturbed estuarine wetlands in Pacific

Northwest, USA - Williams—2025—Ecological Applications—Wiley Online Library. Retrieved March 20, 2026, from <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70011>

Xu, S., Na, M., Miao, Y., Tian, C., Zhou, J., & Liu, X. (2025). Responses of methane emissions to global wetland restoration and influencing factors. *Agricultural and Forest Meteorology*, 364, 110459. <https://doi.org/10.1016/j.agrformet.2025.110459>

Zhang, W., Li, H., Pueppke, S. G., & Pang, J. (2021). Restored riverine wetlands in a headwater stream can simultaneously behave as sinks of N₂O and hotspots of CH₄ production. *Environmental Pollution*, 284, 117114. <https://doi.org/10.1016/j.envpol.2021.117114>